



Monthly Assessment Project

Python for AI & Data Science

Project Title

AI-Based Movie Recommendation System



Project Overview

Recommendation systems are among the most widely used applications of Artificial Intelligence and Data Science in modern digital platforms. Companies such as Netflix, YouTube, and Amazon Prime Video utilize recommendation algorithms to suggest personalized content based on user interests, preferences, and behavior.

In this project, students are required to design and develop a **Beginner-Level AI-Based Movie Recommendation System** using Python programming concepts covered during the course.

The system should recommend movies to users according to:

- Age group
- Preferred movie genre
- Movie ratings
- User interests or mood

This project is intended to strengthen programming logic, data handling skills, and understanding of beginner-level AI concepts.



Project Objectives

The primary objectives of this assessment are to:

- Develop logical and analytical thinking

- Apply Python programming concepts in a real-world scenario
 - Understand the basic workflow of recommendation systems
 - Practice structured data handling using lists and tuples
 - Build a strong foundation for future AI and Data Science projects
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Topics Covered

Students must implement this project using the following concepts only:

- Variables
 - Conditional Statements
 - For Loops
 - Lists
 - Tuples
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Functional Requirements

The Movie Recommendation System must include the following functionalities:

1. Welcome Interface

Display a professional welcome message for the user when the program starts.

Example:

“Welcome to the AI-Based Movie Recommendation System”

2. Movie Dataset Creation

Create and store movie records using lists and tuples.

Each movie record should contain:

- Movie Name

- Genre
- Rating
- Minimum Age Requirement

Example:

("Inception", "Sci-Fi", 9.0, 13)

Students are required to include:

- Minimum 10 movies
 - Minimum 3 different genres
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3. User Input Section

The system should collect the following information from the user:

- User Name
- Age
- Preferred Genre

Optional:

- User Mood or Interest
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4. Recommendation Logic

Implement recommendation logic using conditional statements.

The system should:

- Recommend movies according to the selected genre
- Filter movies according to user age
- Prioritize highly rated movies
- Prevent age-restricted recommendations for underage users

Example:

- Users under 13 should not receive adult movies
- Action lovers should receive action movies
- High-rated movies should be prioritized

✓ 5. Display Recommendations

Display the final recommended movies in a clean and professional format.

Example:

Recommended Movies for Ahmed:

- Interstellar
- Avengers: Endgame
- Spider-Man

✓ 6. Loop Implementation

Students must use for loops to:

- Traverse movie datasets
- Display recommendations
- Count and manage movie records

Mandatory Requirements

Students are required to:

- Use meaningful variable names
- Apply at least 3 conditional statements
- Use loops effectively
- Use lists and tuples appropriately
- Write clean and readable code
- Maintain proper output formatting

Bonus Features (Optional)

Additional marks may be awarded for implementing advanced beginner-level features such as:

- Mood-based movie recommendations
- Multiple genre selection
- Search functionality
- Watch history tracking
- Menu-driven system
- Favorite actor/category filtering

Submission Guidelines

Required Submission Files

Students must submit:

1. Python Source File .py or .ipynb
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File Naming Convention

Format:

RollNo_Name_MovieRecommendationSystem.py

Example:

21_Ahmed_MovieRecommendationSystem.py

Submission Rules

- Code must be properly formatted and readable
- Meaningful variable names should be used
- Program output should be clear and organized
- Plagiarism or copied work will not be accepted
- Late submissions may result in grade deduction